

Polymer-free interfacial graphene transfer

Abstract

This transfer moves a mono- or bi-layer of graphene from its growth substrate (copper) to a new substrate (TEM grid in our case) without using a polymer, solely by using the phase properties of cyclohexane and the chemical properties of APS.

You will find in this document the necessary equipment, the protocol and some remarks, based on our work at the PUCMM nanoscience laboratory.

Our work is based on the article by Liubov et al. *Molecular Caging of Graphene with Cyclohexane: Transfer and Electrical Transport*.

A demonstration of the procedure described in the article is available at this address: <https://www.youtube.com/watch?v=gVoMjUTlsDQ>

We have also produced a video available here: <https://github.com/BASILEzou/STAGE-2026-PUCMM-Diamane/blob/main/transfert%20polymer-free.mp4>

Samples

- Graphene on copper.
- A substrate (SiO₂, TEM grid, glass slide)

Materials

- Two small 20 mL crystallizing dishes
- Three large crystallizing dishes
- Two watch glasses
- One glass funnel
- Four glass slides
- One 100 mL beaker
- Two 5 mL mini beakers
- One 1 mL micropipette
- Teflon tweezers
- PELCO precision tweezers
- A precision spatula
- An ice bath (or the lab's Teflon device)
- Waste containers for APS, acid, and solvents
- Absorbent paper

Major equipment

- A mg scale and weighing paper.
- A fume hood.
- A 2°C refrigerator
- A 0/-1°C freezer
- A sink

Reagents

- 7 mL of 0.5 M APS (0.8 g of ammonium persulfate in 7 mL of water)
- Cyclohexane (a few mL are sufficient)

- Deionized H₂O (DI).
- 2 mL of HNO₃
- Ice cubes (about ten)
- Isopropanol for rinsing equipment

Protective equipment

- Nitrile gloves
- A mask suitable for handling cyclohexane, HNO₃, APS.
- A lab coat
- Safety goggles

Biphasic Transfer Protocol

Preliminary steps (T-2 hours)

Prepare a cooling system: partially fill a large crystallizing dish with DI water and insert a small 20 mL crystallizing dish. The DI water must not overflow. Place this assembly in the freezer.

Place in the freezer, in a large crystallizing dish: two glass slides, Teflon tweezers.

Place in the refrigerator: a large beaker containing DI water (reserve for subsequent steps) and a large crystallizing dish filled with DI water (for rinsing).

Prepare and clean the workstation. Check the operation of the fume hood, refrigerator, freezer, and scale.

Sample preparation

Extract the copper/graphene sheet from the vacuum chamber.

Cut the required area with scissors while holding it with tweezers.

Store the cut section between two glass plates. This will also help flatten the sample. Return the rest of the Cu/graphene to the vacuum chamber.

Backside etching (If the graphene is double-sided)

Put on personal protective equipment.

In a fume hood, in a small crystallizing dish, prepare a solution consisting of 2 mL of HNO₃ and 3 mL of DI water using a 1 mL pipette equipped with specific plastic tips (add the DI water first, then the acid).

Prepare a large crystallizing dish filled with DI water.

Place the sample (floating) on the acid solution using tweezers. Wait 2 minutes (the immersed side becomes silvery, indicating the dissolution of graphene).

Remove the sample with the tweezers, ensuring the top part does not touch the acid, and rinse it by immersion in the large crystallizing dish of DI water.

Dry the sample by capillary action by pressing the edge against a lint-free absorbent paper.

Set the sample aside on a glass slide covered by an inverted 5 mL mini beaker.

Transfer the contaminated solutions into the acid waste container using a funnel, then place the glassware in the sink.

Copper etching

Fold a weighing paper and place it on the scale. Weigh the ammonium persulfate (APS) using a spatula.

Extract the large/small crystallizing dish assembly from the freezer.

Pour the APS into the small crystallizing dish using the spatula under the paper.

In the fume hood, add the DI water with the 1 mL pipette (to achieve a 0.5 M concentration based on the amount of APS).

Place the sample floating on this solution with tweezers.

Add 7 to 8 drops of cyclohexane using the pipette (1 mL) directly onto the sample to cover its entire surface. The addition of these drops must be done gently to apply the least possible mechanical stress on the sample.

Cover the crystallizing dish with a glass cover and place it in the freezer for 15 minutes to ensure uniform cooling.

Take out the crystallizing dishes and add 3 to 4 drops of cyclohexane onto the sample. The cyclohexane should freeze instantly. Return to the freezer for 10 minutes.

Note: The successive additions of cyclohexane compensate for evaporation and allow the graphene to be located post-etching. This also ensures a thick cyclohexane shell around the sample.

Repeat the addition of cyclohexane. Place in the freezer for 5 minutes.

Repeat the addition of cyclohexane. Place the assembly in the refrigerator until complete dissolution of the copper (estimated time of 2 to 4 hours).

Transfer

Once the copper is dissolved, take the crystallizing dish with the sample out of the refrigerator.

Transfer the cyclohexane shell using a glass slide (previously stored in the freezer) into the crystallizing dish of cold DI water that was also in the refrigerator.

Cover and place in the refrigerator for 10 to 15 minutes to wash the cyclohexane shell (which must remain solid and cold).

Meanwhile, take the grid box out of the vacuum chamber, remove one and place it on a cold glass slide, then refrigerate; this allows the grid to adhere well to the slide.

Take the crystallizing dish with the sample out of the refrigerator.

Take the glass slide out of the freezer and place the grid on it a few centimeters from the end.

Using the glass slide holding the grid and the tweezers stored in the freezer, retrieve the cyclohexane, ensuring the area containing the graphene is centered on the grid; the entire cyclohexane shell must fit on the grid.

Place the glass blade at the bottom of the Teflon tray (or ice cube tray), then surround it with ice cubes without them being too close together, then put it in the fridge.

Let the cyclohexane sublime (solid-gas transition) for 20 minutes to 2 hours depending on the thickness.

Observation

When all the cyclohexane has sublimated, remove the glass slide with the substrate on it.

The sample is ready for observation. Handle without applying mechanical stress.

You can observe it under an optical microscope or an electron microscope.

Then, store it in a vacuum box once the observation is complete.

General remarks

All equipment used must be kept very cold at all times.

It is very complicated to control cyclohexane on small surfaces, as it melts and evaporates very quickly.

Also, handling the grid can be complicated at first.

Therefore, do not hesitate to practice beforehand without performing a transfer (for example, cooling cyclohexane and observing how it behaves, handling grids).

Also, do not be discouraged if the transfer does not work; it took me about 10 attempts before getting results.